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Project Name:	Permoda - Módulos Toray MBR 1 X Tren
Project Number:	MX2490 MBR(NHP) 50m3-h 1XTren Rev0

Water Category:	Difficult Industrial Wastewater (e.g. Oil & Gas, Chemical)
MBR System Design	Separate System (the MBR modules are installed in a separate tank)
Calculation of n-1 operation	no

MBR Design (Filtration System) [5.3]

Filtration Flows & Temperature	Unit	N - trains in operation	N-1- trains in operation	Notice
Annual Average Daily Capacity (ADC)	[m ³ /d]	1200	1200	(daily average)
Annual Average Daily Flow (ADF)	[m ³ /h]	50	50	Annual Average Flow: average influent flow in one day without storm water
Hourly Peak Flow (< 1h) (HPF)	[m ³ /h]	0	0	Hourly peak Flow: several hourly peaks are allowed within one day, as long as the average flow of that period is smaller than or equal to the daily peak
Daily Peak Flow (< 24h) (DPF)	[m ³ /h]	0	0	Max. Daily Flow : several daily peaks with in a period of 7 days as long as the average flow of that period is smaller than or equal to the weekly peak
Maximum Daily Capacity	[m ³ /d]	0	0	
Weekly Peak Flow (WPF)	[m ³ /h]	0	0	Maximum Weekly Flow : several weekly peaks with in a period of 30 days as long as the average flow of that period is smaller than or equal to the monthly peak
Monthly Peak Flow (< 30 days) (MPF)	[m ³ /h]	0	0	Maximum Monthly Flow : several monthly peaks are allowed during one year, as long as the average flow of that period is smaller than or equal to the yearly average
Monthly Average Daily Capacity	[m ³ /d]	0	0	(membrane design)
Minimum Temperature in MBR Tank	[°C]	29		(membrane design)

MBR Tanks		
Number of MBR Trains	train(s) / system	1.00
Number of membrane modules per train	module(s) / train	4.00
Number of MBR trains per tank	train(s) / tank	1.00
Number of MBR tanks	tanks	1.00
Basin Length (of each tank)	[m]	5.50
Basin Width (of each tank)	[m]	4.30
Minimum Water Level (m) including 0.5 m superalant	[m]	6.00
Minimum Freeboard	[m]	0.50
Basin Depth	[m]	6.50
Basin Volume (without freeboard)	[m ³]	142
Total Volume of MBR Tank(s)	[m ³]	150

Membrane modules	Unit	N - trains in operation	N-1- trains in operation
Type of membrane modules	type	NHP210-1200DW (future) 304SS	
Total number of membrane modules	pcs	4	0
Number of elements per module	elements / module	1200	
Total number of membrane elements	pcs	4800	0
Installed total membrane surface	[m ²]	3360	0
Flux at Average Daily Flow (ADF)	[L/m ² x h]	14.9	
Flux at Hourly Peak Flow (DPF)	[L/m ² x h]	0.0	
Flux at Daily Peak Flow (DPF)	[L/m ² x h]	0.0	
Flux at weekly peak flow (WPF) (<7 days)	[L/m ² x h]	0.0	
Flux at monthly peak flow (MPF) (< 30days)	[L/m ² x h]	0.0	
Recommended air flow per train	[Nm ³ /h/train]	680	
Maximum air flow per train	[Nm ³ /h/train]	960	
Estimated oxygen input from air scouring (@ rec. flow rate @all train in operation)	[kg/day]	307	
Sludge Recirculation flow per train	[m ³ /h]	167.5	

Design MBR Tank:		
Membrane Module	NHP210-1200DW (future) 304SS	
Membrane Area per Module:	[m ²]	840
Number of Trains (n):	trains	1
Number of Modules per Train:	Modules per Train	4
Number of installed Modules:	Modules	4

Installed Flux:	Unit	Value	Status	Value	Status
Operation Condition		Normal			
Trains in operation	pcs	n			
Number of modules	pcs	4			
Installed Membrane Area	[m ²]	3360			
a) average daily flux	[L/m ² x h]	14.9	O.K.		
b) hourly peak flux	[L/m ² x h]	0.0	n.a.		
c) daily peak flux	[L/m ² x h]	0.0	n.a.		
d) weekly peak flux	[L/m ² x h]	0.0	n.a.		
e) monthly peak	[L/m ² x h]	0.0	n.a.		

MBR related pumps & blowers		
Scouring Air Blower		
recommended scouring air per train	[Nm ³ /h]	680
maximum -design scouring air per train	[Nm ³ /h]	960
no. (excl. piping head loss)	[mbar]	620
MBR Suction Pump(if any)		
required pump capacity per train (rounded)	[m ³ /h]	55.6
max suction head (excl. piping head loss)	[m]	2
MBR Tank Sludge Feed Pump		
Biomass Recirculation MBR Feed	feed by gravity, return by pump	
installed capacity per train	[m ³ /h]	167.5
discharge head	depending on plant design	
Chemical Cleaning (design for one train):		
Volume of Chemical Tank per module	[l]	1080
Volume of Chemical Tank per train	[m ³]	4.32

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